

Crystalline LiPON as a Bulk-Type Solid Electrolyte

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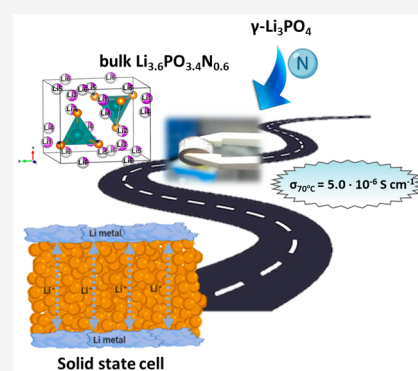


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Supporting Information

ABSTRACT: In this work, a new bulk $\text{Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ crystalline polymorph has been prepared from low-cost precursors, following a simple ball-milling procedure. The densified powder exhibits a conductivity of $5.0 \times 10^{-6} \text{ S cm}^{-1}$ at 70°C and an electrochemical stability allowing operation with high-voltage materials up to 5.0 V vs Li/Li^+ . Stripping and plating of lithium in a symmetric cell demonstrates the forthcoming bulk application of LiPON in electrochemical devices. Widening the use of lithium phosphorus oxynitride compositions to bulk solid-state batteries will have relevant implications because of its unique compatibility with both high-voltage electroactive materials and lithium metal and its low density.



Lithium phosphorus oxynitride (LiPON) compositions first received significant attention in the 1990s as solid electrolyte (SEs) for the development of thin-film solid-state batteries (SSBs).^{1,2} LiPONs are commonly prepared via sputtering of Li_3PO_4 in a N_2 atmosphere, resulting in amorphous glass thin films with a wide range of compositions $\text{Li}_x\text{P}_y\text{N}_z$ ($2.6 \leq x \leq 3.5$, $1.9 \leq y \leq 3.8$, $0.1 \leq z \leq 1.3$).^{3–5} Unlike other SEs, sputtered *s*-LiPON (where the *s* denotes “sputtered”) possesses desirable characteristics.⁶ First, LiPONs are less sensitive to air than other solid electrolytes, such as $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO) garnet or sulfides. Second, in contrast to $\text{Li}_{1-x}\text{Al}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ (LATP) or sulfide-based electrolytes, which are easily reduced in contact with Li and form unfavorable solid electrolyte interfaces (SEI),^{8–10} a stable and favorable SEI is formed between Li metal and LiPON as revealed by recent studies.¹¹ In addition, LiPON is one of the few stable solid-electrolyte materials that has enabled the fabrication of anodeless batteries.^{12,13} Indeed, excellent electrochemical results have been reported in thin-film batteries using *s*-LiPONs as the SE, in combination with positive active materials such as TiS_2 , LiMn_2O_4 , and LiCoO_2 .^{14,15} However, these devices do not exceed ca. 1 cm^2 with thicknesses of $10 \text{ }\mu\text{m}$,^{14,15} because of its limited ionic conductivity ($\sim 2 \times 10^{-6} \text{ S cm}^{-1}$ at room temperature),⁶ and they operate at low current densities (on the order of tens of $\mu\text{A cm}^{-2}$). Moreover, the limitations of the sputtering technique hamper upscaling toward cells delivering capacities of hundreds of milliampere-hours or larger. Therefore, the synthesis of a stable bulk *b*-LiPON (where *b* denotes “bulk”) remains critical for the transition toward its application in SSBs.

Ceramic SEs are one of the most promising approaches to build safe, high-powered, and high-energy density batteries, although this premise relies on the use of a Li-metal anode^{7,10} paired with high-voltage cathode materials such as layered oxides. The specific weight of the SE material has a critical impact on the energy density of the final device. A SE thickness of $<50 \text{ }\mu\text{m}$ is required to provide competitive volumetric and gravimetric energy densities (E_v , E_g).¹⁶ Indeed, electrolytes including heavy metals (such as LLZO) must compensate for their density ($\sim 4.8 \text{ g cm}^{-3}$) with a thickness of $<20 \text{ }\mu\text{m}$ ¹⁷ in order to attain competitive energy density, which, currently, is technologically challenging. Therefore, the low density of LiPONs ($\sim 2.4 \text{ g cm}^{-3}$) provides considerable motivation for its development as bulk SE material. Figure 1 compares the expected theoretical gravimetric and volumetric energy densities (E_g and E_v , respectively) of high-voltage Li-metal SSBs, including either a *b*-LiPON, a ceramic LLZO, or a glass-ceramic LATP as SEs. The cell models consider a thickness of $50 \text{ }\mu\text{m}$. The cathode composition includes 75 wt % of

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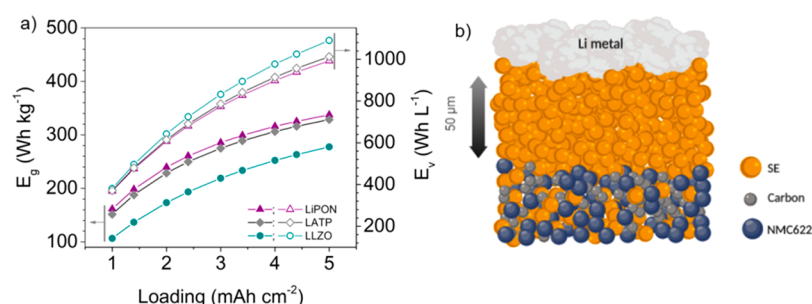


Figure 1. (a) Expected theoretical gravimetric and volumetric energy densities (E_g and E_v , respectively) of high-voltage Li-metal SSBs, as a function of the cathode loading and including LiPON, LATP, or LLZO as SEs. (b) Schematic representation of the cell model considered for the theoretical predictions shown in panel (a).

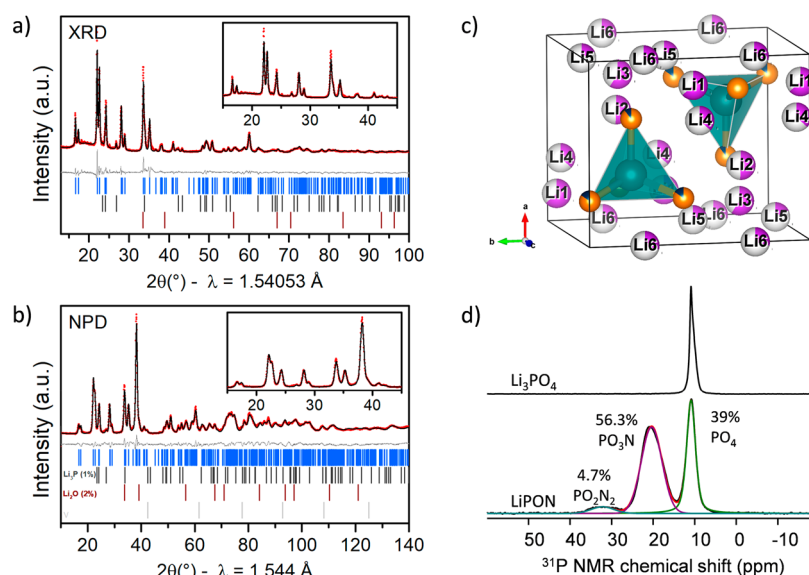


Figure 2. Results of the combined Rietveld refinement of (a) the XRD pattern and (b) the NPD pattern of $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$. (c) Representation of the refined structure. The six Li positions are shown as purple and white balls, indicating the partial occupation of each site. Positions of P, O, and N are shown as green, orange, and blue balls, respectively, forming tetrahedral cationic sites drawn in green pointing in opposite directions (up–down pattern). (d) ^7Li and ^{31}P solid-state MAS NMR spectra of $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$. One average signal is observed for the Li, while three distinct environments are clearly observed for the P. The attribution and quantification of these three fitted peaks are indicated.

Table 1. Crystal Structure Data of $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$, Obtained from the Combined Rietveld Refinement of the XRD and NPD Patterns

space group: $P2_1/m$		$\rho = 2.38 \text{ g cm}^{-3}$		$\chi^2 = 13.4$	
XRD pattern		$V = 167.13(19) \text{ \AA}^3$		$R_{\text{Bragg}} = 13.5\%$	
$a = 5.098(4) \text{ \AA}$	$b = 6.158(4) \text{ \AA}$	$c = 5.324(4) \text{ \AA}$		$\beta = 90.60(4) \text{ \AA}^3$	
NPD pattern		$V = 166.68(7) \text{ \AA}^3$		$R_{\text{Bragg}} = 5.1\%$	
$a = 5.094(2) \text{ \AA}$	$b = 6.150(2) \text{ \AA}$	$c = 5.321(2) \text{ \AA}$		$\beta = 90.53(2) \text{ \AA}^3$	
atom	Wyckoff site	x/a	y/b	z/c	occupancy
P1	2e	0.334(3)	3/4	0.677(4)	1.0
O1/N1	4f	0.220(2)	0.538(2)	0.809(3)	0.84(5)/0.16(5)
O2/N2	2e	0.363(3)	1/4	0.310(4)	0.89(3)/0.11(3)
O3	2e	0.230(4)	3/4	0.396(3)	1.0
Li1	4f	0.173(9)	0.505(10)	0.183(13)	0.62(5)
Li2	2e	0.36(2)	1/4	0.66(3)	0.46(4)
Li3	2e	0.173(19)	1/4	0.637(18)	0.54(4)
Li4	4f	0.362(16)	0.531(15)	0.16(2)	0.38(5)
Li5	4f	0.06(4)	0.07(3)	1/2	0.19(4)
Li6	2e	0	1/4	0	0.29(5)

NMC622 active material, 5 wt % of carbon C65, and 20 wt % of SE. Table S1 in the Supporting Information summarizes all the parameters taken into account for the calculations. E_v ranges from 200 Wh l^{-1} to ca. 1000 Wh l^{-1} for cathode loading between 1 and 5 mAh cm^{-2} , without relevant differences between the electrolytes. However, pronounced differences are observed for the trend of E_g : in the best scenario, with a cathode loading of 5 mAh cm^{-2} , LiPON and LATP light electrolytes lead to E_g values as high as 320 Wh kg^{-1} (i.e., 60% higher than for the heavier LLZO). Indeed, according to our calculations, the thickness of the LLZO electrolyte should be reduced down to the challenging value of 30 μm in order to reach similar E_g values. Such low thicknesses, as well as the high sintering temperature, seriously challenge the processing at a large scale.^{18,19}

To the best of our knowledge, only a couple of crystalline forms of LiPON have been reported so far: $\text{Li}_{0.88}\text{PO}_{3.73}\text{N}_{0.14}$ ²⁰ and $\text{Li}_2\text{PO}_2\text{N}$;²¹ both of them with limited ionic conductivity at room temperature ($\sim 1 \times 10^{-13}$ S cm^{-1}). While these two phases were obtained from solid-state reactions under a controlled atmosphere at high temperatures (600 and 950 $^{\circ}\text{C}$, respectively), in this work, crystalline *b*-LiPON was prepared through simple mechanical milling of the low-cost precursors LiPO_3 and Li_3N for 20 min under Ar (see the Supporting Information for further details). SEM images of the as-synthesized *b*-LiPON (Figures S1a and S1b in the Supporting Information) revealed particles with irregularly shaped morphology and sizes ranging from 0.1 μm to 1 μm , and with some forming agglomerates.

The powder X-ray diffraction (XRD) pattern of *b*-LiPON could be indexed in a monoclinic lattice, with the space group $P2_1/m$ and cell parameters $a = 5.098(3)$ $\text{\AA}$, $b = 6.158(4)$ $\text{\AA}$, $c = 5.324(4)$ $\text{\AA}$, and $\beta = 90.60(4)^{\circ}$. The structure of the new *b*-LiPON phase was first determined from neutron powder diffraction (NPD) data, using $\text{Li}_{3.75}\text{Si}_{0.75}\text{P}_{0.25}\text{O}_4$ ²² as the starting structural model, before being confirmed with a combined refinement of both NPD and XRD patterns (see the Supporting Information for further details on the structure determination). The results of these refinements are presented in Figures 2a and 2b, and in Table 1. The *b*-LiPON structure is built on a slightly distorted hexagonal close packing (hcp) array of O and N anions. The N anions were found to partially occupy two of the three crystallographically independent oxygen sites O1/N1 and O2/N2. P cations occupy $\text{PO}_{3.57(13)}\text{N}_{0.43(13)}$ coordination tetrahedra, which are isolated from the others and are alternately pointing in opposite directions (in a regular up–down pattern; see Figure 2c). Li cations were found in six distinct positions, with the same coordination environments as in the Li_4SiO_4 parent structure: Li6 is found in an octahedral coordination, Li5 is 5-fold coordinated, while the other four Li occupy tetrahedral sites. The individual occupancies of the six Li sites sum 3.67(13), and the overall refined composition is in agreement with the chemical composition $\text{Li}_{3.6(1)}\text{PO}_{3.4(1)}\text{N}_{0.6(1)}$ reproducibly determined on several samples from complementary energy-dispersive X-ray (EDX) spectroscopy, inductively coupled plasma optical emission spectrometry (ICP-OES), and elemental analysis (EA) (see Table S3 in the Supporting Information).

The thermal stability of the *b*- $\text{Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ powder under air was assessed by simultaneous thermogravimetric (TGA), differential thermal (DTA), and mass spectrometry (MS) analyses (see Figure S2a in the Supporting Information). The

results evidence an exothermic decomposition reaction with the irreversible loss of N occurring between 450 $^{\circ}\text{C}$ and 650 $^{\circ}\text{C}$. The weight gain observed in the TGA is in agreement with the oxidation reaction reported elsewhere.²¹ The XRD pattern of the powder after heating at 600 $^{\circ}\text{C}$ reveals the decomposition of *b*- $\text{Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ to form Li_3PO_4 (Figure S2b in the Supporting Information). The XRD of the pellet of *b*- $\text{Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ sintered at 350 $^{\circ}\text{C}$ under air (Figure S2b) confirms the stability the *b*- $\text{Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ phase for the pellets prepared at this temperature for the ionic conductivity measurements presented below. On the other hand, after having observed that *b*- $\text{Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ reacts with water (not shown), we observed that, while *b*- $\text{Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ evolves in a few days to form Li_2CO_3 when exposed to ambient air, the XRD pattern of the sample remains unchanged under dry atmosphere after 5 days (see Figure S3 in the Supporting Information).

The ^7Li solid-state NMR spectrum of *b*- $\text{Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ shows a single resonance with a single exponential longitudinal relaxation time (see Figure 2d, as well as Figure S4b in the Supporting Information). This result suggests that all of the different Li sites are rapidly exchanging positions in the NMR time scale. Three different signals are observed in the ^{31}P NMR spectrum (Figure 2d), which are assigned to the three different environments of phosphorus: PO_4 (4.7%), PO_3N (56.3%), and PO_2N_2 (39%). The two-dimensional (2D) ^{31}P – ^{31}P -EXSY correlation obtained with a mixing time of 2 s (Figure S4c in the Supporting Information) shows correlation peaks between all different lines, confirming the spatial proximity of all the different phosphorus sites and the single-phase nature of the sample.

Extensive studies show that the insertion of nitrogen into Li_3PO_4 thin films to obtain compositions such as $\text{Li}_{2.88}\text{PO}_{3.73}\text{N}_{0.14}$ raises the conductivity from 7×10^{-8} S cm^{-1} to 2×10^{-6} S cm^{-1} at room temperature (RT)^{1,23–25} and improves the electrochemical stability of the electrolyte up to 5.5 V vs Li/Li^+ .^{1,26,27} The nitridation of lithium phosphate glasses modifies the phosphate anion chains and forms –N= linked phosphate anion chains.²⁶ Despite the recent significant progress, how the N is incorporated into the phosphate structure and how it correlates with the ionic conductivity still remains a subject of research.²⁷ Some studies propose that the increase of the ionic conductivity after the nitridation arises from the increasing structural cross-linking, and decreasing electrostatic energy after replacing the P–O bond by the more covalent P–N one.²⁶ This bonding has been confirmed in *b*- $\text{Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ by the ^{31}P NMR solid-state spectra shown in Figure 2d.

The ionic conductivity of *b*- $\text{Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ was experimentally measured in order to confirm such enhancement, with respect to the phosphate analogue. Densification of SEs in SSBs is a key step that is required to promote not only high ionic conductivity, but a low-resistive interface with lithium metal, as well as to mitigate the dendrite propagation. The high-pressure and low-temperature (HPLT) sintering technique is particularly suitable for powders exhibiting limited thermal stability, as is the case for our *b*- $\text{Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$.^{28,29} Pellets ca. 1 mm thick were prepared from the as-synthesized powder quasi-isostatically pressed at 5.5 GPa for 2 min at temperatures ranging from 200 $^{\circ}\text{C}$ to 450 $^{\circ}\text{C}$, under ambient atmosphere. A *b*- $\text{Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ pellet was also prepared at 1.25 GPa at RT for comparison. The main characteristics of the pellets are summarized in Table S2 in the Supporting

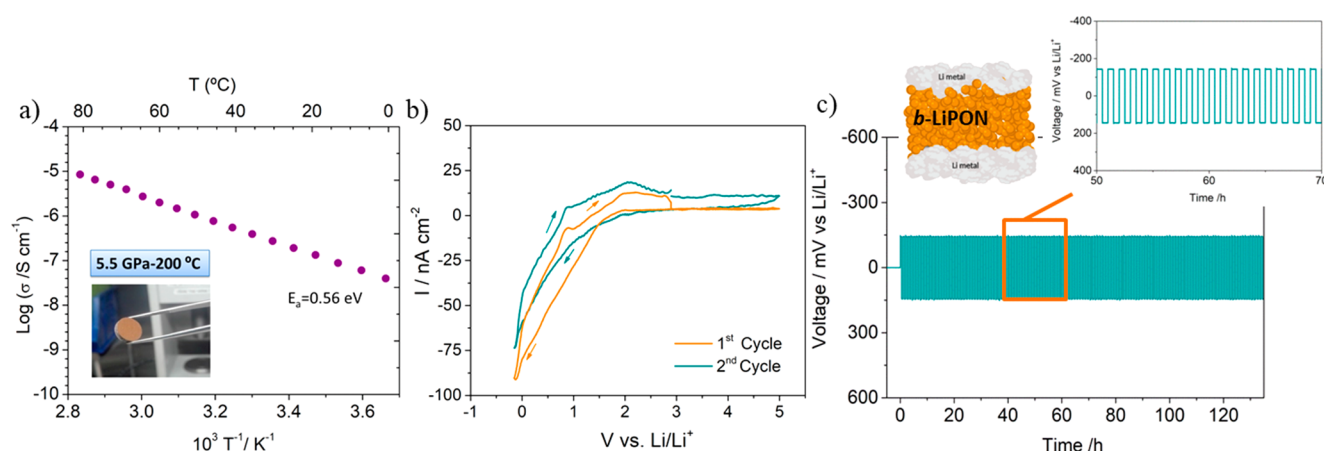


Figure 3. (a) Ionic conductivity as a function of the temperature for a $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ pellet prepared by HPLT at 5.5 GPa and 200 °C. (b) Cyclic voltammetry of $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ acquired at 0.5 mV s^{-1} in the -0.15 – 5 V vs Li^+/Li^0 for two cycles. (c) Stripping and plating experiment of a symmetric $\text{Li}|b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}|\text{Li}$ cell at 70 °C (inset shows data between 50 h and 70 h).

Information. The latter compact exhibited a relatively low density of ca. 75% (the theoretical density deduced from the unit-cell volume is 2.38 g cm^{-3}). In contrast, highly dense pellets (87%–95%) were obtained via HPLT sintering, and no clear trend was found for the relative density when increasing the sintering temperature. Indeed, the highest density among the samples explored (95%) was obtained at 250 °C. The morphology of the surface and cross section of the HPLT bodies observed by SEM (Figures S1e and S1f in the Supporting Information) shows a fully and uniformly densified structure. Electrochemical impedance spectroscopy (EIS) measurements were performed on symmetric Li cells of the HPLT pellets ($\text{Li}|b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}|\text{Li}$) (Figure S5a in the Supporting Information). The temperature-dependent conductivity studied from 0 to 80 °C is represented in the Arrhenius plot of Figure 3a. The conductivity of $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ pelletized at low pressure and at 25 °C (75% dense) reached $5.6 \times 10^{-8} \text{ S cm}^{-1}$ at 25 °C. This value increased up to 1 order of magnitude for the pellets prepared by HPLT (Figure S5b in the Supporting Information) and reached $6.5 \times 10^{-6} \text{ S cm}^{-1}$ at 70 °C for the pellet pressed at 5.5 GPa and 200 °C (90% dense, Figure 3a). The superior conductivity of HPLT processed pellets is likely due to the higher density attained with this technique, which proves fully suitable for low-temperature densification of ceramics. A similar conductivity was found for samples prepared at lower temperatures (see Table S2), without significant trends associated with the applied sintering temperature (see Figure S5b). Although it still remains insufficient to operate in a full cell at room temperature ($\sim 10^{-4} \text{ S cm}^{-1}$ required),³⁰ the progress made in this work with $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ is remarkable if compared to $s\text{-LiPONs}$ ($\sim 10^{-6}$ – $10^{-5} \text{ S cm}^{-1}$ at RT and 70 °C),⁶ and well above that of $\gamma\text{-Li}_3\text{PO}_4$ ($4.2 \times 10^{-18} \text{ S cm}^{-1}$ at RT) and the other bulk phosphorus oxynitrides reported so far: $\text{Li}_{0.88}\text{PO}_{3.73}\text{N}_{0.14}$ ($1.4 \times 10^{-13} \text{ S cm}^{-1}$ at RT)³¹ and $\text{Li}_2\text{PO}_2\text{N}$ ($8.8 \times 10^{-7} \text{ S cm}^{-1}$ at 80 °C).²¹

The average activation energy for $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ is 0.55 eV (Table S2), similar to the previously reported crystalline $\text{Li}_2\text{PO}_2\text{N}$ (0.57 eV)²¹ and to that reported for amorphous $s\text{-LiPON}$ thin films. The electrical insulating property of $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$, a sine qua non feature for any electrolyte, was confirmed by chronoamperometry (Figure S6 in the Supporting Information).

Cells prepared with $s\text{-LiPONs}$ have been reported to exhibit a wide stability window (0 to 5 V vs Li/Li^+).^{14,15} Several publications claimed a successful chemical stabilization of the interface between the Li anode and the electrolyte induced by the addition of a thin film of $s\text{-LiPON}$ buffer interlayer.^{32–34} Indeed, a stable SEI including Li_2O , Li_3N , and Li_3PO_4 has been recently unveiled between lithium metal and LiPON .¹¹ On the cathode side, Put et al. reported that, although $s\text{-LiPONs}$ decomposes above 4.5 V vs Li/Li^+ , the decomposition products ($\text{Li}_4\text{P}_2\text{O}_7$, LiPO_3 , and P_4O_{10}) are stable at higher potentials and that the layer formed at the interface shields the electrolyte.³⁵ This explains the excellent cyclability of SSBs using $s\text{-LiPON}$ and a high-voltage AM.³⁶ In this work, the electrochemical stability of the HPLT $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ pellets was evaluated by cyclic voltammetry using Li metal disk and silver as reference and working electrodes, respectively. Voltammograms were acquired at 70 °C in the -0.15 – 5 V range vs Li^0/Li^+ , applying a scan rate of 0.5 mV s^{-1} . The cyclic voltammetry in Figure 3b shows a reductive current below 100 nA cm^{-2} , attributed to the lithium deposition occurring below 0 V. Besides, it demonstrates the excellent electrochemical stability for $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ up to 5 V, confirmed by the extremely low oxidative current below 12 nA cm^{-2} arising from the residual current of the partial electronic conductivity of the solid electrolyte (see Figure S6).

In Li-metal SSBs, the high interface resistance appearing between Li^0 and the SE is a major drawback for the battery performance.³⁷ A symmetric $\text{Li}|b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}|\text{Li}$ cell showed a stable ASR of $4 \times 10^5 \Omega \text{ cm}^2$ at RT after conditioning the cell at 70 °C for 24 h. Interestingly, this resistance has been found to be similar to the Li/LLZO garnet interface,³⁸ showing a comparable limited effective contact area between the electrolyte and the electrodes. The stripping-plating abilities of $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ were examined with the symmetric lithium metal cell at 70 °C (see the Supporting Information for experimental details). The investigation of lithium dendrite nucleation and growth resistance in the cell was performed using prolonged galvanostatic cycling tests (Figure 3c). The SE exhibits an impressive cycling performance with 35 cycles (140 h) without dendrite propagation leading to a short circuit. The current density induced an overpotential of ca. 180 mV in both directions, with a constant voltage indicating a homogeneous dissolution and deposition of metallic Li during the cycling.³⁹

These results highlight the outstanding progress toward a real bulk full cell including a lithium phosphorus oxynitride composition as SE.

In conclusion, we report the facile synthesis of a bulk crystalline $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$ solid electrolyte, which may be prepared via mechano-synthesis from the low-cost precursors LiPO_3 and Li_3N . A high-pressure (5.5 GPa) and low-temperature (200–400 °C) technique is presented here as an interesting approach to obtain pellets 23% denser than the powder pelletized at 1.25 GPa. These highly dense pellets exhibit a conductivity of $3.0 \times 10^{-7} \text{ S cm}^{-1}$ at 25 °C, which increases to $5.0 \times 10^{-6} \text{ S cm}^{-1}$ at 70 °C. The wide electrochemical stability of this electrolyte makes it suitable for use with high-voltage active materials, and the excellent stripping-plating performance of lithium in a symmetric $\text{Li}|\text{b-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}|\text{Li}$ cell is shown here as a proof of concept that approaches a real electrochemical device. Therefore, this work represents a remarkable step forward for the transition from LiPON thin films toward bulk application in the field of solid-state batteries.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acseenergylett.0c02336>.

Parameters for theoretical calculations, experimental methods, and additional figures (PDF)

Crystal structure of $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$, as determined via NPD (CIF)

Crystal structure of $b\text{-Li}_{3.6}\text{PO}_{3.4}\text{N}_{0.6}$, as determined by XRD (CIF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Yu, X.; Bates, J. B.; Jellison, G. E.; Hart, F. X. A Stable Thin-Film Lithium Electrolyte: Lithium Phosphorus Oxynitride. *J. Electrochem. Soc.* **1997**, *144* (2), 524–532.
- (2) Bates, J. B.; Dudney, N. J.; Gruzalski, G. R.; Zuhur, R. A.; Choudhury, A.; Luck, C. F.; Robertson, J. D. Fabrication and Characterization of Amorphous Lithium Electrolyte Thin Films and Rechargeable Thin-Film Batteries. *J. Power Sources* **1993**, *43* (1), 103–110.
- (3) Bates, J. B.; Dudney, N. J.; Gruzalski, G. R.; Zuhur, R. A.; Choudhury, A.; Luck, C. F.; Robertson, J. D. Electrical Properties of Amorphous Lithium Electrolyte Thin Films. *Solid State Ionics* **1992**, *53–56*, 647–654.
- (4) Oudenhoven, J. F. M.; Baggetto, L.; Notten, P. H. L. All-Solid-State Lithium-Ion Microbatteries: A Review of Various Three-Dimensional Concepts. *Adv. Energy Mater.* **2011**, *1* (1), 10–33.
- (5) Nowak, S.; Berkemeier, F.; Schmitz, G. Ultra-Thin LiPON Films – Fundamental Properties and Application in Solid State Thin Film Model Batteries. *J. Power Sources* **2015**, *275*, 144–150.
- (6) Zheng, F.; Kotobuki, M.; Song, S.; Lai, M. O.; Lu, L. Review on Solid Electrolytes for All-Solid-State Lithium-Ion Batteries. *J. Power Sources* **2018**, *389*, 198–213.
- (7) Albertus, P.; Babinec, S.; Litzelman, S.; Newman, A. Status and Challenges in Enabling the Lithium Metal Electrode for High-Energy and Low-Cost Rechargeable Batteries. *Nat. Energy* **2018**, *3* (1), 16–21.
- (8) Gao, Z.; Sun, H.; Fu, L.; Ye, F.; Zhang, Y.; Luo, W.; Huang, Y. Promises, Challenges, and Recent Progress of Inorganic Solid-State Electrolytes for All-Solid-State Lithium Batteries. *Adv. Mater.* **2018**, *30*, 1870122.
- (9) Schnell, J.; Günther, T.; Knoche, T.; Vieider, C.; Köhler, L.; Just, A.; Keller, M.; Passerini, S.; Reinhart, G. All-Solid-State Lithium-Ion and Lithium Metal Batteries – Paving the Way to Large-Scale Production. *J. Power Sources* **2018**, *382*, 160–175.
- (10) Janek, J.; Zeier, W. G. A Solid Future for Battery Development. *Nature Energy* **2016**, *1*, 16141–16144.
- (11) Cheng, D.; Wynn, T. A.; Wang, X.; Wang, S.; Zhang, M.; Shimizu, R.; Bai, S.; Nguyen, H.; Fang, C.; Kim, M.; Li, W.; Lu, B.; Kim, S. J.; Meng, Y. S. Unveiling the Stable Nature of the Solid

Electrolyte Interphase between Lithium Metal and LiPON via Cryogenic Electron Microscopy. *Joule* **2020**, 4 (11), P2484–P2500.

(12) Wang, M. J.; Carmona, E.; Gupta, A.; Albertus, P.; Sakamoto, J. Enabling “Lithium-Free” Manufacturing of Pure Lithium Metal Solid-State Batteries through in Situ Plating. *Nat. Commun.* **2020**, 11 (1), 5201–5209.

(13) Neudecker, B. J.; Dudney, N. J.; Bates, J. B. Lithium-Free” Thin-Film Battery with In Situ Plated Li Anode. *J. Electrochem. Soc.* **2000**, 147 (2), 517–523.

(14) Bates, J. B.; Dudney, N. J.; Neudecker, B.; Ueda, A.; Evans, C. D. Thin-Film Lithium and Lithium-Ion Batteries. *Solid State Ionics* **2000**, 135 (1), 33–45.

(15) Patil, A.; Patil, V.; Wook Shin, D.; Choi, J.-W.; Paik, D.-S.; Yoon, S.-J. Issue and Challenges Facing Rechargeable Thin Film Lithium Batteries. *Mater. Res. Bull.* **2008**, 43 (8), 1913–1942.

(16) McCloskey, B. D. Attainable Gravimetric and Volumetric Energy Density of Li–S and Li Ion Battery Cells with Solid Separator-Protected Li Metal Anodes. *J. Phys. Chem. Lett.* **2015**, 6 (22), 4581–4588.

(17) Gutiérrez-Pardo, A.; Pitillas Martinez, A. I.; Otaegui, L.; Schneider, M.; Roters, A.; Llordés, A.; Aguesse, F.; Buannic, L. Will the Competitive Future of Solid State Li Metal Batteries Rely on a Ceramic or a Composite Electrolyte? *Sustainable Energy Fuels* **2018**, 2 (10), 2325–2334.

(18) Pfenninger, R.; Struzik, M.; Garbayo, I.; Stilp, E.; Rupp, J. L. M. A Low Ride on Processing Temperature for Fast Lithium Conduction in Garnet Solid-State Battery Films. *Nat. Energy* **2019**, 4 (6), 475–483.

(19) Chan, C.; Yang, T.; Weller, J. M. Nanostructured Garnet-type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$: Synthesis, Properties, and Opportunities as Electrolytes for Li-ion Batteries. *Electrochim. Acta* **2017**, 253, 268–280.

(20) Wang, B.; Chakoumakos, B. C.; Sales, B. C.; Kwak, B. S.; Bates, J. B. Synthesis, Crystal Structure, and Ionic Conductivity of a Polycrystalline Lithium Phosphorus Oxynitride with the $\gamma\text{-Li}_3\text{PO}_4$ Structure. *J. Solid State Chem.* **1995**, 115 (2), 313–323.

(21) Senevirathne, K.; Day, C. S.; Gross, M. D.; Lachgar, A.; Holzwarth, N. A. W. A New Crystalline LiPON Electrolyte: Synthesis, Properties, and Electronic Structure. *Solid State Ionics* **2013**, 233, 95–101.

(22) Baur, W. H.; Ohta, T. The Crystal Structure of $\text{Li}_{3.75}\text{Si}_{0.75}\text{P}_{0.25}\text{O}_4$ and Ionic Conductivity in Tetrahedral Structures. *J. Solid State Chem.* **1982**, 44 (1), 50–59.

(23) Ayu, N. I. P.; Kartini, E.; Prayogi, L. D.; Faisal, M.; Supardi. Crystal Structure Analysis of Li_3PO_4 Powder Prepared by Wet Chemical Reaction and Solid-State Reaction by Using X-Ray Diffraction (XRD). *Ionics* **2016**, 22 (7), 1051–1057.

(24) Lee, C.; Dutta, P. K.; Ramamoorthy, R.; Akbar, S. A. Mixed Ionic and Electronic Conduction in Li_3PO_4 Electrolyte for a CO_2 Gas Sensor. *J. Electrochem. Soc.* **2006**, 153 (1), H4–H14.

(25) Takada, K. Progress and Prospective of Solid-State Lithium Batteries. *Acta Mater.* **2013**, 61 (3), 759–770.

(26) Wang, B.; Kwak, B. S.; Sales, B. C.; Bates, J. B. Ionic Conductivities and Structure of Lithium Phosphorus Oxynitride Glasses. *J. Non-Cryst. Solids* **1995**, 183 (3), 297–306.

(27) Lacivita, V.; Westover, A. S.; Kercher, A.; Phillip, N. D.; Yang, G.; Veith, G.; Ceder, G.; Dudney, N. J. Resolving the Amorphous Structure of Lithium Phosphorus Oxynitride (Lipon). *J. Am. Chem. Soc.* **2018**, 140 (35), 11029–11038.

(28) Urbanovich, V. S. Computerized System for the Sintering of Nanoceramics at High Pressures. *Powder Metall. Met. Ceram.* **2003**, 42 (1), 19–23.

(29) Galceran, M.; Pujol, M. C.; Gluchowski, P.; Stręk, W.; Carvajal, J. J.; Mateos, X.; Aguiló, M.; Díaz, F. A Promising $\text{Lu}_{2-x}\text{Ho}_x\text{O}_3$ Laser Nanoceramic: Synthesis and Characterization. *J. Am. Ceram. Soc.* **2010**, 93 (11), 3764–3772.

(30) Inada, R.; Yasuda, S.; Tojo, M.; Tsuritani, T.; Tojo, T.; Sakurai, Y. Development of Lithium-Stuffed Garnet-Type Oxide Solid Electrolytes with High Ionic Conductivity for Application to All-Solid-State Batteries. *Front. Energy Res.* **2016**, 4, 28.

(31) Wang, B.; Chakoumakos, B. C.; Sales, B. C.; Kwak, B. S.; Bates, J. B. Synthesis, Crystal Structure, and Ionic Conductivity of a Polycrystalline Lithium Phosphorus Oxynitride with the $\gamma\text{-Li}_3\text{PO}_4$ Structure. *J. Solid State Chem.* **1995**, 115 (2), 313–323.

(32) West, W. C.; Whitacre, J. F.; Lim, J. R. Chemical Stability Enhancement of Lithium Conducting Solid Electrolyte Plates Using Sputtered LiPON Thin Films. *J. Power Sources* **2004**, 126 (1), 134–138.

(33) Dudney, N. J. Addition of a Thin-Film Inorganic Solid Electrolyte (Lipon) as a Protective Film in Lithium Batteries with a Liquid Electrolyte. *J. Power Sources* **2000**, 89 (2), 176–179.

(34) Visco, S. J.; Chu, M.-Y. Protective Coatings for Negative Electrodes. U.S. Patent No. US6025094A, Feb. 15, 2000.

(35) Put, B.; Vereecken, P. M.; Stesmans, A. On the Chemistry and Electrochemistry of LiPON Breakdown. *J. Mater. Chem. A* **2018**, 6 (11), 4848–4859.

(36) Li, J.; Ma, C.; Chi, M.; Liang, C.; Dudney, N. J. Solid Electrolyte: The Key for High-Voltage Lithium Batteries. *Adv. Energy Mater.* **2015**, 5 (4), 1401408–1401416.

(37) Famprikis, T.; Canepa, P.; Dawson, J. A.; Islam, M. S.; Masquelier, C. Fundamentals of Inorganic Solid-State Electrolytes for Batteries. *Nat. Mater.* **2019**, 18, 1278–1291.

(38) Alexander, G. V.; Patra, S.; Sobhan Raj, S. V.; Sugumar, M. K.; Ud Din, M. M.; Murugan, R. Electrodes-Electrolyte Interfacial Engineering for Realizing Room Temperature Lithium Metal Battery Based on Garnet Structured Solid Fast Li^+ Conductors. *J. Power Sources* **2018**, 396, 764–773.

(39) Tsai, C.-L.; Roddatis, V.; Chandran, C. V.; Ma, Q.; Uhlenbruck, S.; Bram, M.; Heitjans, P.; Guillon, O. $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ Interface Modification for Li Dendrite Prevention. *ACS Appl. Mater. Interfaces* **2016**, 8 (16), 10617–10626.